Unit 8, Lesson 10: Surface Area of Rectangular Prisms

Benchmark

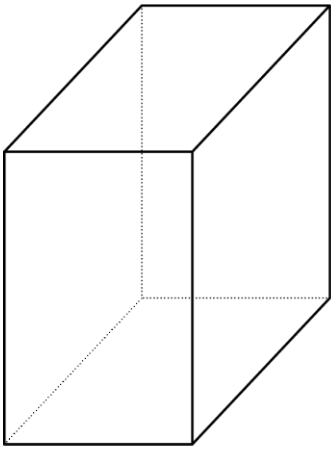
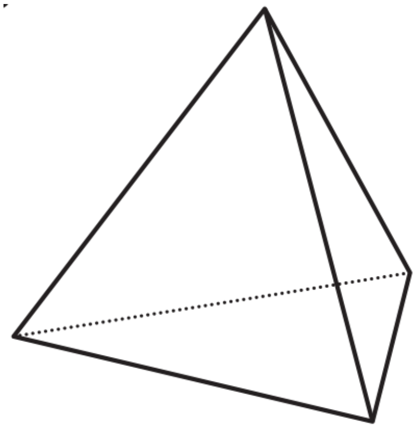
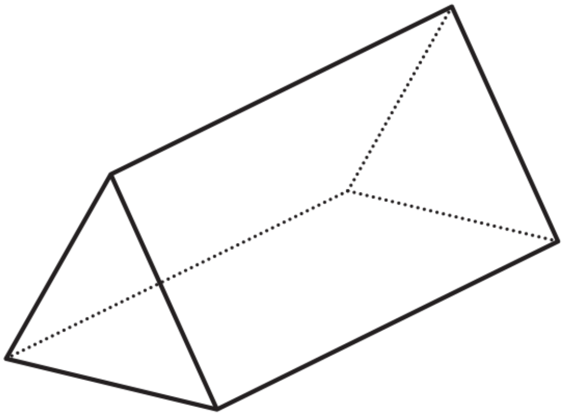
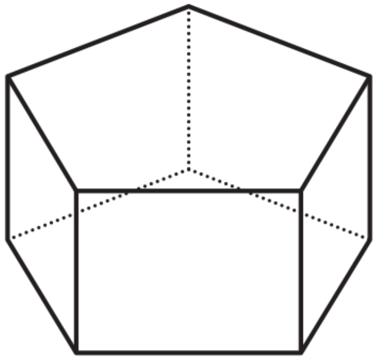
MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure’s net.

Learning Target

☐ I can find the surface area of a rectangular prism using a net or the figure.

8.10.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.

A	B
	
C	**D**
	

a. Which one doesn't belong?

b. Justify your choice using mathematics.

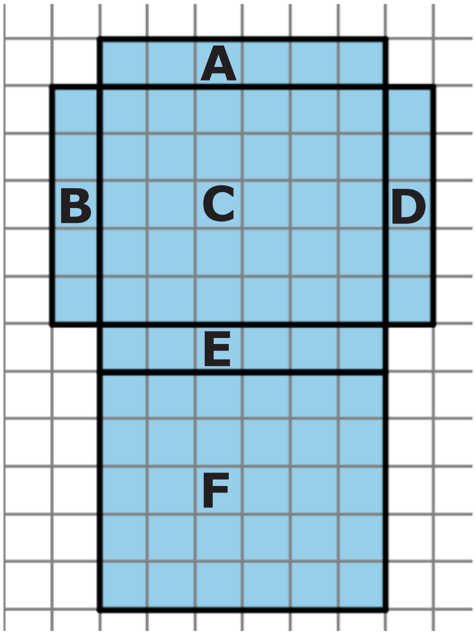
8.10.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

Surface area is the total area of the two-dimensional surfaces that make up a three-dimensional object

1. A net of a rectangular prism is shown below with each face labeled.

a. Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



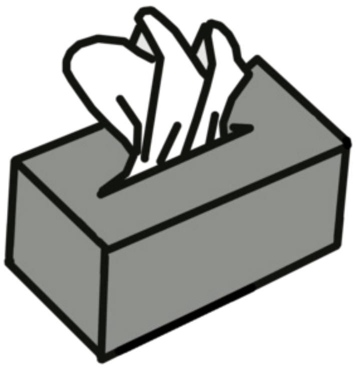
Face	Dimensions	Area
A		
B		
C		
D		
E		
F		

b. How can the net and the area of the six faces be used to find the surface area of the prism?

c. What is the surface area of this rectangular prism?

When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



a. Sketch the net of the tissue box.

b. Label the dimensions of each face on the net.

c. Find the area of each face. Include units.

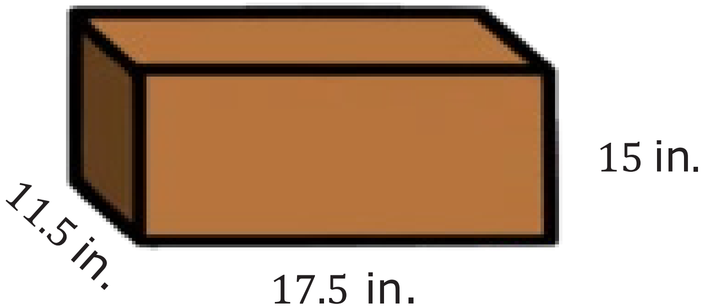
d. Add the area of the faces to find the surface area of the tissue box. Include units.

8.10.3 Guided Practice: Surface Area of a Rectangular Prism

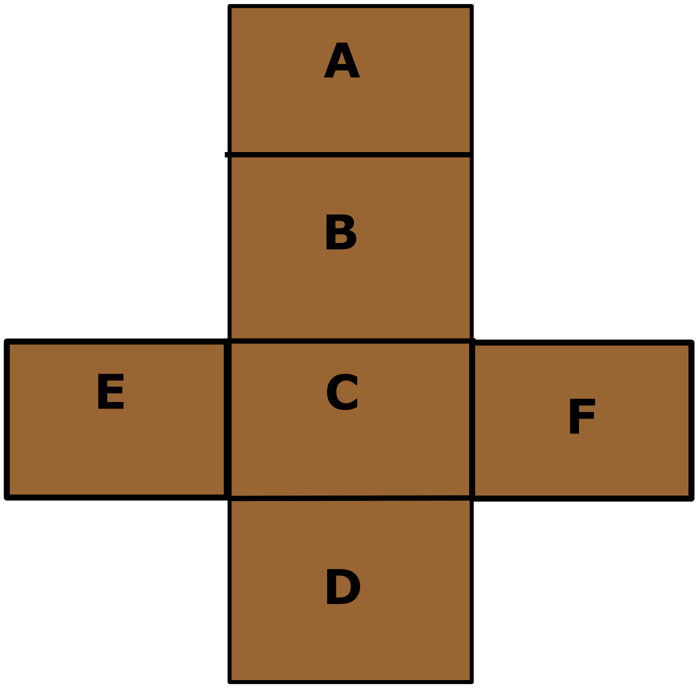
1. Gretchen is preparing to paint a jewelry box but does not know how much paint she needs to buy.

    a. What information about the jewelry box does Gretchen need to know in order to buy paint?

    b. Gretchen measured the box to be 17.5 inches (in.) long, 11.5 in. wide, and 15 in. tall.



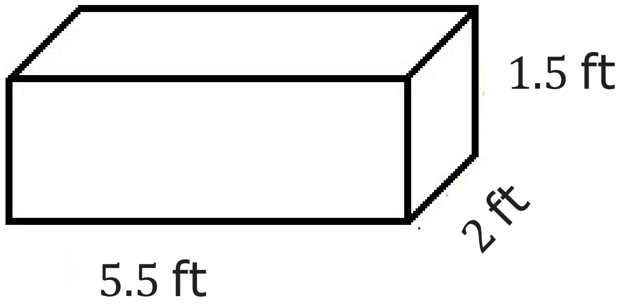
The net of the jewelry box is shown. Label the side lengths of the net. Record the dimensions in the table.



Face	Dimensions	Area
A		
B		
C		
D		
E		
F		

c. What is the surface area of the box that Gretchen needs to buy paint for? Include units.

2. Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.

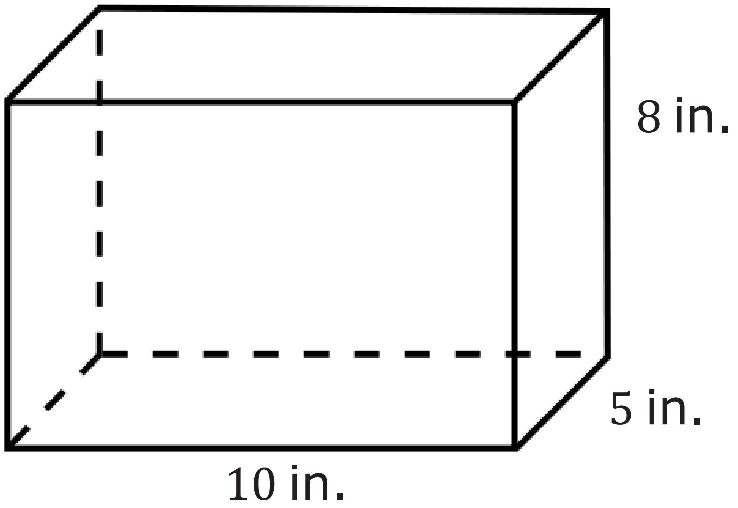


Jillian's Work	Victor's Work
Side 1: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$ Side 2: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$ Side 3: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$ Side 4: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$ Side 5: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$ Side 6: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$ Total Surface Area: 44.5 ft^2	2 sides: $2(5.5 \text{ ft} \times 1.5 \text{ ft}) = 16.5 \text{ ft}^2$ 2 sides: $2(5.5 \text{ ft} \times 2 \text{ ft}) = 22 \text{ ft}^2$ 2 sides: $2(2 \text{ ft} \times 1.5 \text{ ft}) = 6 \text{ ft}^2$ Total Surface Area: 44.5 ft^2

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.

8.10.4 Your Turn!

1. Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?

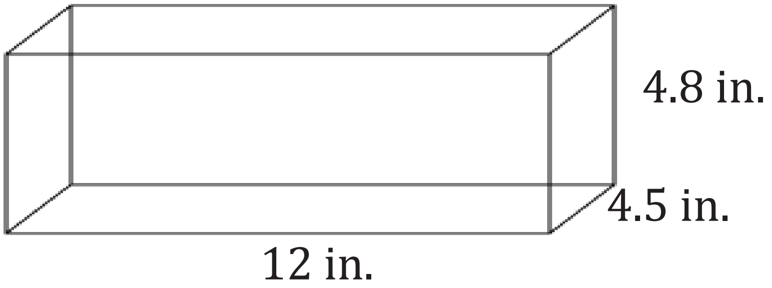


2. A carpenter is staining planks of wood to build the top of a long picnic table. The carpenter determines the surface area of each plank in order to purchase the right amount of stain. Each plank is $\frac{3}{4}$ inches (in.) thick, $5\frac{1}{2}$ in. wide, and 72 in. long. What is the surface area of each plank?

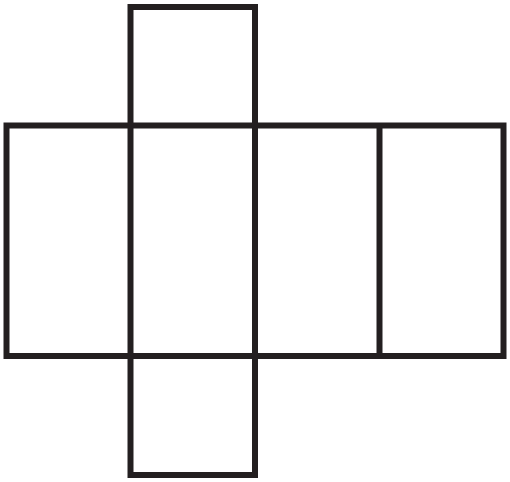
8.10.5 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



1. Write the measurements on the net of the box.



2. Use the net to find the surface area of the shipping box. Include units.